

SDK3.0 Document Overview

The device ID with D in the middle is the engineering card, and it does not support secondary development if it is not an engineering card

Note: Look at the title header, make sure it is a engineering card and then look at others

1. SDK3.0 The protocol is compatible with SDK2.0 documents, and the connection process is the same.
2. The maximum length of a TCP packet is 9×1024 bytes. When the xml length exceeds $9 \times 1024 - 4$, it needs to be divided into packets.
3. All SDK data is transmitted in UTF-8 encoding formatSDK
4. Request information command values are **kSDKCmdAsk**
5. This document is not the final version. If there is a new function and it is not in the document information, please update to the latest document. If the latest document does not have this information, you can apply to add relevant documents to the document.

1.Protocol

The device ID with D in the middle is the engineering card, and it does not support secondary development if it is not an engineering card

Note: Look at the title header, make sure it is a engineering card and then look at others

Steps required for SDK connection

1. Make Tcp connection.
2. Conduct connection version negotiation
3. Carry out SDK negotiation (GetIFVersion), in order to obtain the guid used for subsequent interactions
4. The SDK negotiation has been completed, and the xml data interaction will be performed according to the SDK protocol
5. SDK protocol data uses little-endian byte order
6. SDK interface data uses UTF-8 encoding format

Ask SDK related questions

1. First of all, if it is a functional problem, you should first use the SDK test tool to verify whether it is a device problem.
2. If it is a device problem, you need to inform the firmware version number of the device when asking the question. It can be obtained through HDPlayer or GetDeviceInfo of the SDK.
3. If it is not a device problem, then you also need to explain its version number when asking questions, and tell how to reproduce the problem and the content sent. Let the person who solves the problem know the reproduction process. Then this problem will soon enter the processing stage , instead of asking the solver to continuously test and reproduce the problem, which will delay the resolution time of the problem.

1.1 Basic type definition

```
typedef unsigned long long    hint64;
typedef unsigned int         hint32;
typedef unsigned short       hint16;
typedef unsigned char        hint8;

typedef long long            hint64;
typedef int                  hint32;
typedef short                hint16;
typedef char                 hint8;
typedef float                hfloat;
typedef double               hdouble;

#define MAX_DEVICE_ID_LENGTH    15
#define MD5_LENGTH              32
#define LOCAL_TCP_VERSION      0x1000007
#define LOCAL_UDP_VERSION      0x1000007
```

1.2 Error code definition

The error code return format is as follows:

Field	Command packet length (Len)	Command(Cmd)	Error Code(Code)
Bytes	2 bytes	2 bytes	2 byte
Value	6	kErrorAnswer	HErrorCode

HErrorCode definition

The value increases sequentially from 0, such as normal state 0, next write file completed 1, process error 2

```
enum HErrorCode
{
    kSuccess = 0,           ///< Normal status
    kWriteFinish,          ///< Write file complete
    kProcessError,         ///< Processor error
    kVersionTooLow,        ///< Version too low
    kDeviceOccupa,         ///< Device is occupied
    kFileOccupa,           ///< File is occupied
    kReadFileExcessive,    ///< Excessive users of read back file
}
```

kInvalidPacketLen,	///< Data packet length error
kInvalidParam,	///< Invalid parameter
kNotSpaceToSave,	///< Not enough storage space
kCreateFileFailed,	///< Create file failed
kWriteFileFailed,	///< Write file failed
kReadFileFailed,	///< Read file failed
kInvalidFileData,	///< Invalid file data
kFileContentError,	///< File content error
kOpenFileFailed,	///< Open file error
kSeekFileFailed,	///< Seek file error
kRenameFailed,	///< Rename failed
kFileNotFound,	///< File not found
kFileNotFinish,	///< File received not finish
kXmlCmdTooLong,	///< xml command is too long
kInvalidXmlIndex,	///< Invalid Xml command index value
kParseXmlFailed,	///< Parsing Xml error
kInvalidMethod,	///< Invalid Method name
kMemoryFailed,	///< Memory failed
kSystemError,	///< System error
kUnsupportedVideo,	///< Unsupported video
kNotMediaFile,	///< Not media file
kParseVideoFailed,	///< Parsing video file failed
kUnsupportedFrameRate,	///< Unsupported baud
kUnsupportedResolution,	///< Unsupported resolution(video)
kUnsupportedFormat,	///< Unsupported format(video)
kUnsupportedDuration,	///< Unsupported time length(video)
kDownloadFileFailed,	///< Download file failed
kScreenNodeIsNull,	///< Screen node is null
kNodeExist,	///< Node has already exist
kNodeNotExist,	///< Node is not exist
kPluginNotExist,	///< Plugin not exist
kCheckLicenseFailed,	///< Check license failed
kNotFoundWifiModule,	///< Not found wifi module
kTestWifiUnsuccessful,	///< Test wifi module not successful
kRunningError,	///< Running error
kUnsupportedMethod,	///< Unsupported method
kInvalidGUID,	///< Invalid guid
kFirmwareFormatError,	///< Firmware format error
kTagNotFound,	///< Tag not exist
kAttrNotFound,	///< Attribute does not exist
kCreateTagFailed,	///< Create tag failed
kUnsupportedDeviceType,	///< Unsupported device type

};

1.3 Data packet command value definition

```
enum HcmdType
{
    kTcpHeartbeatAsk      = 0x005f,    ///< TCP heartbeat packet request
    kTcpHeartbeatAnswer   = 0x0060,    ///< TCP heartbeat packet feedback
    kSearchDeviceAsk      = 0x1001,    ///< Search device request
    kSearchDeviceAnswer   = 0x1002,    ///< Search device response
    kErrorAnswer          = 0x2000,    ///< Error answer
    kSDKServiceAsk        = 0x2001,    ///< version negotiation request
    kSDKServiceAnswer     = 0x2002,    ///< version negotiation response
    kSDKCmdAsk            = 0x2003,    ///< sdkcommand request
    kSDKCmdAnswer         = 0x2004,    ///< sdk command feedback
    kGPSInfoAnswer        = 0x3007,    ///< gps information response
    kFileStartAsk         = 0x8001,    ///< File start transfer request
    kFileStartAnswer      = 0x8002,    ///< File start transfer response
    kFileContentAsk       = 0x8003,    ///< Requests with file content
    kFileContentAnswer    = 0x8004,    ///< Response to write file content
    kFileEndAsk           = 0x8005,    ///< File end transfer request
    kFileEndAnswer        = 0x8006,    ///< File end transfer response
    kReadFileAsk          = 0x8007,    ///< Read back file request
    kReadFileAnswer       = 0x8008,    ///< Read back file response
};
```

1.4 Protocol packet structure format

Protocol packets are mainly used for:

1. Send and read back files
2. SDK request and feedback in Xml format

Field	Command packet length (len)	Command(cmd)
Bytes	2 bytes	2 bytes

```
#pragma pack(1)
typedef struct HTcpHeader
{
```

```

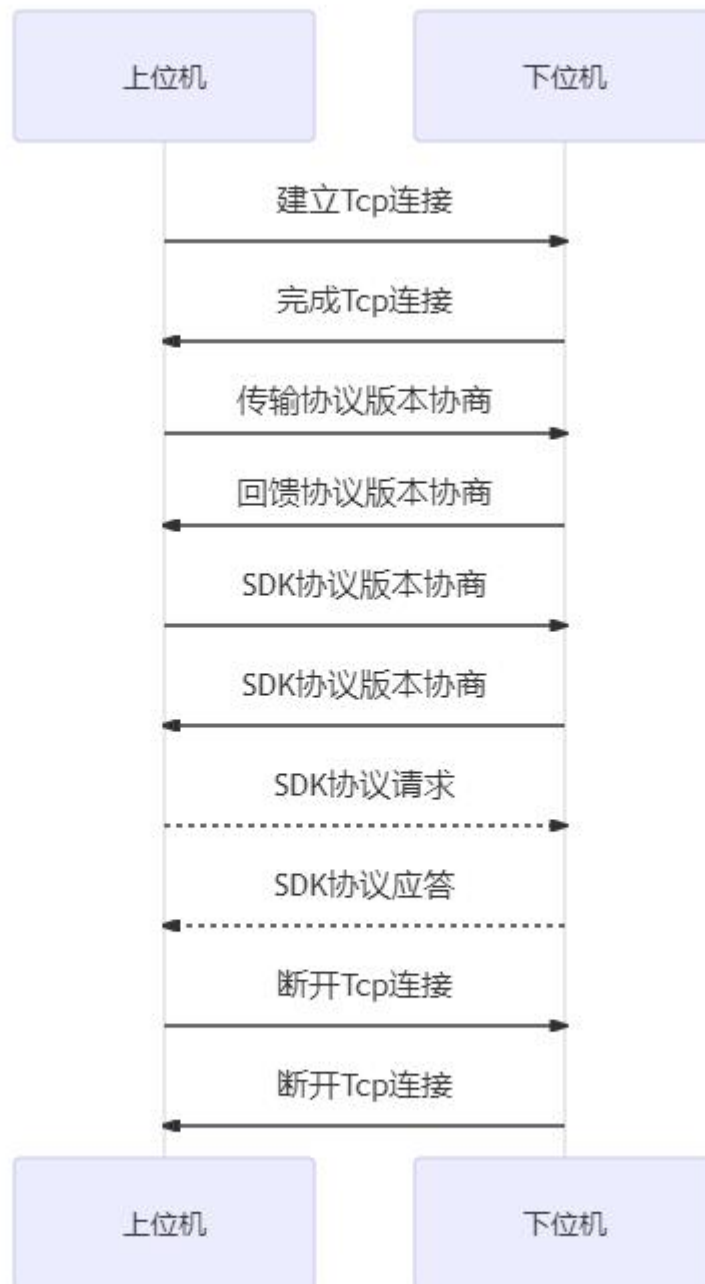
    uint16 len;    ///< Command packet length
    uint16 cmd;    ///< Command length value
} HTcpHeader;

typedef struct HTcpExt
{
    uint16 len;    ///< Command packet length
    uint16 cmd;    ///< Command value
    uint8  ext[0]; ///< Extended data start address
} HTcpExtAsk, HTcpExtAnswer;
#pragma pack()

```

1.4 SDK Protocol process

1. First establish a Tcp connection
2. Interact with the protocol version
3. For SDK information interaction, when there is no SDK information interaction for a period of time, heartbeat packets need to be exchanged to maintain the connection
4. Close the Tcp connection after no further SDK interaction



Version negotiation

1. The upper computer sends a version negotiation packet, the version number is LOCAL_TCP_VERSION
2. After the lower computer receives the version negotiation package, it extracts the version number,,

- If the version number is less than the minimum version number supported by the lower computer, return the error message that the transmission protocol version is too low
 - If the version number is greater than the highest version number supported by the lower computer, return the highest version number of the lower computer, and the upper computer sets the highest version number returned by the lower computer
 - If the version number is in the lower computer support list, the lower computer will automatically switch to the same transmission protocol and return the same version number
3. 3. Received the error message that the transmission version is too low because the connection is disconnected

Protocol version request

Field	Command packet length(Len)	Command(Cmd)	Version number(Version)
Bytes	2 bytes	2 bytes	4 bytes
Value	8	kSDKServiceAsk	LOCAL_TCP_VERSION

Protocol version feedback

Field	Command packet length(Len)	Command(Cmd)	Version number(Version)
Bytes	2 bytes	2 bytes	4 bytes
Value	8	kSDKServiceAsk	Feedback value

SDK version negotiation

The SDK protocol version negotiation obtains the guid value through GetIFVersion SDK, and the subsequent SDK communication of this connection needs to use the guid value to complete the SDK request

Before the SDK protocol interaction, you need to obtain the guid value, which needs to be filled in the guid of the xml sdk tag in the subsequent SDK interaction.

Note: The SDK version negotiation uses the SDK protocol interaction, and the SDK protocol headers are the same, as shown in the format of the SDK request below.

SDK protocol interaction

The SDK data packet protocol depends on the protocol data packet structure, and the maximum length of each packet cannot exceed the protocol data packet requirements

SDK request

Field	Command packet length(Len)	Command(Cmd)	Total length	Index	SDK interface data
Bytes	2 bytes	2 bytes	4 bytes	4 bytes	N byte
Value	8	kSDKServiceAsk	Total	index	XmlData

Len value: $Len = 2 + 2 + 4 + 4 + N$;

Total value is the total length of SDK interface data.

The index value is the index of the SDK interface data of the current package. For example, when the data is larger than the protocol data package, it needs to be divided into packages. The first package is 0, and the second package is the xml data + the offset position of the sent xml data. For example, sending 200 bytes Xml, divided into 1 package of 100 bytes, then the Index of the first package = 0, and the Index of the second package = 100;

The XmlData value is the SDK interface data, the maximum length requirement and the protocol packet length, $XmlMaxLength = 9\ 1024 - 2 - 2 - 4 - 4$ bytes, which is $9\ 1027$ - the length of the packet header.

SDK Response

Field	Command packet length(Len)	Command(Cmd)	Total length	Index	SDK interface data
Bytes	2 bytes	2 bytes	4 bytes	4 bytes	N byte
Value	8	kSDKServiceAsk	Total	index	XmlData

Error code feedback format

Field	Command packet length(Len)	Command(Cmd)	Error code(Code)
Bytes	2 bytes	2 bytes	2 bytes
Value	8	kSDKServiceAsk	HErrCode

1.6 SDK version negotiation

The SDK interface data used for the first SDK protocol version negotiation, because the guid value is not yet clear.

In the currently connected SDK communication, the guid attribute in the sdk tag must be the guid value returned by the SDK version negotiation.

SDK version negotiation request

Requirement :

1. After negotiating the transmission protocol version, it is necessary to send the following SDK version request to the lower computer for SDK version negotiation
2. In this SDK, only the value of the value attribute is variable, and the others are fixed values. Now use 1000000

```
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="##GUID">
  <in method="GetIFVersion">
    <version value="1000000"/>
  </in>
</sdk>
```

SDK version negotiation feedback

Feedback information:

1. The value of the guid attribute of the sdk tag, this value is the current connection for subsequent SDK protocol interaction, and needs to be recorded and kept
2. The value of the result attribute of the out tag. This value indicates the status value. For example, kSuccess indicates the normal status. For details, see the error code definition

```
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="f7b3b451c4f3cf3310e2d6714fbb50e8">
  <out method="GetIFVersion" result="kSuccess">
    <version value="1000000"/>
  </out>
</sdk>
```

2、Introduce the structure of program list in xml

examples:

```
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
  <in method="AddProgram">
    <screen timeStamps="0">
      <program guid="##program-guid" type="normal">
        <backgroundMusic/>
        <playControl count="1" disabled="false"/>
        <area alpha="255" guid="##area-guid">
          <rectangle x="0" height="128" width="128" y="0"/>
          <resources>
            <text guid="text-guid" singleLine="false">
              <style valign="middle" align="center"/>
              <string>例子</string>
              <font name="SimSun" italic="false" bold="false"
underline="false" size="28" color="#ffffff"/>
            </text>
          </resources>
        </area>
      </program>
    </screen>
  </in>
</sdk>
```

2.1 Program list data

The program list is composed of xml, and its program tags are sub-tags related to SDK programs

SDK Related tags of program

```
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
  <in method="##Program">
    <!-- The beginning here is the programlist tag data -->
  </in>
</sdk>
```

The structure of program list tag

Program nodes, area nodes, etc. These nodes have some necessary attributes

```
<!-- Program list sub-tag structure, used to explain the program list structure, does not
include SDK tags, attributes, etc. -->
<!--Screen node-->
<screen>
  <!-- program node, allowing multiple -->
  <program>
    <!--area node, noticed this area is sub-area of the program, allowing multiple-->
    <area>
      <!-- Area range, xy position and width and height of nodes in this area -->
      <rectangle x="##x" height="##height" width="##width" y="##y"/>
      <!-- Resource node, its sub-nodes represent the program type and data, and
its sub-nodes allow multiple -->
      <resources>
        <!-- The subtags here represent programs, such as text, video, image,
clock etc, allowing multiple program tags-->
        </resources>
      </area>
    </program>
  </screen>

<!--program Node required attribute
##guid The unique identifier of the program, if the value is empty, the specified program
cannot be updated and deleted
##type normal: Indicates a normal program; template: indicates a template program; html5:
indicates an HDML5 template program; default value: normal
-->
<!--area Node required attribute
##guid The unique identifier of area
-->
```

2.2 Program Tag Attributes

screen node attributes

```
<!--screen node attributes description
```

##timeStamps Indicates the timestamp of screen creation. When the value is different from the last time, all programs on the current device will be cleared.

It can be the total number of seconds from January 01, 1970 00:00:00 GMT to the present.

-->

```
<screen timeStamps="##timeStamps">
  <program>...</program>
</screen>
```

program node attributes

Notes: It is necessary for guid attributes, this attribute distinguishes the unique identifier of the show node. Can not be common.

Exist the selected sub-node

border the tag of border

backgroundMusic Tag of background music

playControl Tag of play control

<!--program Node attribute description

##guid The unique identifier of the program, if the value is empty, the specified program cannot be updated and deleted

##type Node type{normal: Indicates a normal program; template: indicates a template program; html5: indicates an HDML5 template program; default value: normal

##flag The special label of the program {realtime: indicates a real-time program, when there is a program with this label, only the program will be played; default value: empty}

-->

```
<program guid="##guid" type="##type" flag="##flag">
  <!--selected sub-tag-->
  <border/>
  <backgroundMusic/>
  <playControl/>
  <area>...</area>
</program>
```

border Program sub-node attributes

<!--border border function description

##index The serial number of the border pattern (starting from 0) The value refers to the border function of HDPlayer

##effect Animation effect of the border { rotate: rotate (default); blink: blink; null: still } (optional)

```
##speed Border speed { slow: slow; middle: moderate (default); fast: fast } (optional)
-->
<border index="##index" effect="##effect" speed="##speed"/>
```

backgroundMusic Program sub-node attributes

```
<!--backgroundMusic Background music description
##name File name(Allowing mutiple)
-->
<backgroundMusic>
    <file name="##name"/>
</backgroundMusic>
```

playControl Program sub-node attributes

```
<!--playControl Play control tag
##duration Indicates the fixed duration of the program {hh:mm:ss (default: the time to play
the program once)} (optional, mutually exclusive with count)
##count The number of times the program is played continuously {[0-999] default: 1}
(optional, mutually exclusive with duration)
##disabled Whether the program is disabled, if the program is disabled, the program will not
play {"true", "false" (default)}
##week.enable Weeks allowed to play { Mon, Tue, Wed, Thur, Fri, Sat, Sun (multiple use ",
to separate) }
##date.start Start date { YYYY-MM-DD }
##date.end End date{ YYYY-MM-DD }

##time.start Start date { hh:mm:ss (24 hour clock) }
##time.end End date { hh:mm:ss (24 hour clock) }

##location.lng longitude { Double type}
##location.lat latitude { Double type }
##location.range Range, the unit is kilometers { Double type }
##location.preemptive Preemptive play, when entering the specified area,
If the currently playing program is not fixed-point or not in the current area, stop the current
program immediately and switch to this program{"true", "false"(default)}

##location.playOutside Whether to allow outside play {"true"(default), "false"}
-->
<playControl duration="##duration" count="##count" disabled="##disabled">
    <!-- optional-->
    <week enable="##enable"/>
    <!-- optional, allow mutiple-->
```

```

<date start="##start" end="##end"/>
<time start="##start" end="##end"/>
<location lng="##lng" lat="##lat" range="##range" preemptive="##preemptive"
playOutside="##playOutside"/>
</playControl>

```

area Node attributes

Note: There must be a guid attribute, which distinguishes the unique identifier of the area node. It cannot be the same:

When there are multiple regions, when a program exceeds a certain number of regions (approximately about 10), and the stay time is short, there may be a flickering phenomenon. At this time, please set the stay time duration of the regions that do not need to be turned as much as possible. If the value is higher, the exact value should be set only when the page needs to be turned, so as to avoid the phenomenon of flickering.

```

<!--area Node attribute description
##guid Unique identifier for the region
##alpha Transparency of the area {0~255 0 fully transparent 255 opaque; default: 255}
##rectangle.x The x-axis where the region is positioned on the show
##rectangle.y The y-axis where the region is positioned on the show
##rectangle.width The area occupies the width of the program ##rectangle.height The area
occupies the height of the program
-->
<area guid="##guid" alpha="##alpha">
  <rectangle x="##x" height="##height" width="##width" y="##y"/>
  <!--resourcesTags have no attributes, and their sub-tags are program tags-->
  <resources>...</resources>
  <!--optional sub-tag-->
  <border/>
</area>

```

resources Node attributes

Tags have no attributes, and their sub-tags are program tags

2.3 Program node attributes

Belongs to the child nodes of the resource node (resources)

text Text program

```
<!--  
##text.guid Unique ID of the text  
##text.singleLine Whether it is a single line of text {"true" (default), "false"}  
##text.background Background color, if there is no such attribute, it will be a transparent  
color  
String The node is the content of the text program  
  
style font effect The node is a function node (optional). For details, please go to Function  
Node Properties-->  
<text guid="##guid" singleLine="##singleLine" background="##background">  
  <style valign="##valign" align="##align" />  
  <string>文本内容</string>  
  <font name="##name" italic="##italic" bold="##bold" underline="##underline"  
size="##size" color="##color" />  
  <effect inSpeed="##inSpeed" duration="##duration" in="##in"  
outSpeed="##outSpeed" out="##out" />  
</text>
```

Image Image program

```
<!--  
##image.guid Unique ID of the image  
##fit The fill property of the image {  
  fill: For filling, first enlarge the picture to a ratio that can cover the entire area, and then  
  cut out the middle part for display.      center: Center, shrink the image proportionally to the  
  size of the area, and black borders will be displayed when the ratio is inconsistent  
  stretch: Stretching, which may cause image distortion (default)      tile : tiling  
}  
##file.name file name  
file effect The node is a function node (optional). For details, please go to Function Node  
Properties  
-->  
<image guid="##guid" fit="##fit">  
  <file name="name" />  
  <effect inSpeed="##inSpeed" duration="##duration" in="##in"  
outSpeed="##outSpeed" out="##out"/>  
</image>
```

Video Video program

```
<!--  
##video.guid Unique ID of the video  
##video.aspectRatio keep aspect ratio {"true", "false" (default)}  
##file.name file name  
  
file The node is a function node (optional). For details, please go to Function Node  
Properties-->  
<video guid="##guid" aspectRatio="##aspectRatio">  
  <file name="name" />  
</video>
```

clock Clock program

```
<!--  
##clock.guid Unique ID of the clock  
##clock.type { digital: digital clock dial: analog clock } (optional)  
##clock.timezone Time zone { In the format of "+8:00"; default value: automatically adapt to  
the system time zone } (optional)  
##clock.adjust Time fine-tuning { "+00:05:00" forward time; "-00:05:00" backward time  
Default: 0 } (optional)  
##clock.lang Translation language, default zh_cn (optional) { "zh_cn", "zh_tw", "en", "tr",  
"de", "es", "fa",  
"ja", "ko", "ru", "pl", "id", "ar", "it", "fr", "el", "ms", "pt", "sr", "th", "vi" }  
  
{  
Optional font settings, different types of clock configurations:  
  dial Add font child nodes directly to the clock node, or add attributes to the  
corresponding nodes, giving priority to the attributes of the corresponding nodes  
  digital The attribute value needs to be added to the corresponding node, the  
fontSize attribute can change the font size, and the fontName attribute can change the font}  
Child nodes are optional  
##title.value title content  
  
##date.format date format, enter number {  
0、YYYY/MM/DD  
1、MM/DD/YYYY  
2、DD/MM/YYYY  
3、Jan DD, YYYY  
4、DD Jan, YYYY  
5. MM, DD, YYYY (translated language)  
6. DD on MM (translated language)
```

```

}

##week.format The format of the week, enter number {
0. Monday (translated language)1、 Monday
2、 Mon
}

##time.format time format, input number {
0. hh:mm:ss
1. hh:ss
2. hh hour mm minute ss second (translated language)
3. hh hour mm minute (translated language)
}

Clock Options Common
##color color in the format "#RRGGBB"
##display Whether to display {"true" (default), "false"}-->
<clock guid="##guid" type="##type" timezone="##timezone" adjust="##adjust"
lang="##lang">
  <title value="##value" color="##color" display="##display"/>
  <date format="##format" color="##color" display="##display"/>
  <week format="##format" color="##color" display="##display"/>
  <time format="##format" color="##color" display="##display"/>
  <lunarCalendar color="##color" display="##display"/>
</clock>

```

table Table program

The table itself supports persistent data. If you need to restore the default data, you need to set it to true on the clearData attribute of the table tag

```

<!--
##table.guid Unique ID of the table
##table.transparent background transparent {"true", "false" (default)}
##table.clearData clear data {"true", "false" (default)}
##table.row number of rows
##table.col number of columns

##effect... Effect node, please go to function node properties to view

##border.color The color of the line of each cell
##border.lineWidth The width of the line of each cell

Uniform attribute: row means [0, ...) row col means [0, ...) column

```

##size.width size width indicates the width of the specified column

##size.height size height indicates the height of the specified row

Merge cells, if the merged content has been merged, then it will not be merged

The method of unmerging, specifying the same rows and columns that have been merged, and unmerging when two values are equal

For example, unmerge the properties of the merger: row == mergeRow; col == mergeCol

##merger.mergeRow which column to merge into

##merger.mergeCol which line to merge into

cell content

##background.value Default "#000000", background color

##background.file background image, when the background image is set, the background color is invalid

##text cell content (choose one)

##file cell file (currently only supports pictures, choose one)

-->

```
<table guid="##guid" transparent="##transparent" clearData="##clearData" row="##row" col="##col">
```

```
  <effect inSpeed="##inSpeed" duration="##duration" in="##in" outSpeed="##outSpeed" out="##out"/>
```

```
  <border color="##color" lineWidth="##lineWidth"/>
```

```
  <size row="##row" col="##col" width="##width" height="##height"/>
```

```
  <merger row="##row" col="##col" mergerRow="##mergerRow" mergerCol="mergerCol"/>
```

```
  <cell row="##row" col="col">
```

```
    <style valign="##valign" align="##align" />
```

```
    <font name="##name" italic="##italic" bold="##bold" underline="##underline" size="##size" color="##color" />
```

<!--When the value exists, there is no need to fill in the file-->

```
value="##value">
```

```
  <file .../>
```

```
</background>
```

<background

<!--The following text or file (currently only supports pictures) can choose one-->

```
<text>...</text>
```

```
  <file .../>
```

```
</cell>
```

```
</table>
```

2.4 Function node attributes

Description of the attribute of the function node attached to the program node

Effect node

Effect type

```
enum EffectType
{
IMMEDIATE_SHOW = 0, ///< display directly.
    LEFT_PARALLEL_MOVE = 1, ///< Pan to the left.
    RIGHT_PARALLEL_MOVE = 2, ///< Pan to the right.
    UP_PARALLEL_MOVE = 3, ///< Pan up.
    DOWN_PARALLEL_MOVE = 4, ///< Pan down.
    LEFT_COVER = 5, ///< cover to the left.
    RIGHT_COVER = 6, ///< cover to the right.
    UP_COVER = 7, ///< cover up.
    DOWN_COVER = 8, ///< cover down.
    LEFT_TOP_COVER = 9, ///< top left cover.
    LEFT_BOTTOM_COVER = 10, ///< bottom left cover.
    RIGHT_TOP_COVER = 11, ///< top right cover.
    RIGHT_BOTTOM_COVER = 12, ///< bottom right cover.
    HORIZONTAL_DIVIDE = 13, ///< Horizontal split.
    VERTICAL_DIVIDE = 14, ///< vertical split.
    HORIZONTAL_CLOSE = 15, ///< Horizontal close.
    VERTICAL_CLOSE = 16, ///< vertical close.
    FADE = 17, ///< fade in and out.
    HORIZONTAL_SHUTTER = 18, ///< Horizontal shutter.
    VERTICAL_SHUTTER = 19, ///< vertical shutter.
    NOT_CLEAR_AREA = 20, ///< does not clear the screen.
    LEFT_SERIES_MOVE = 21, ///< move left continuously.
    RIGHT_SERIES_MOVE = 22, ///< move right continuously.
    UP_SERIES_MOVE = 23, ///< move up continuously.
    DOWN_SERIES_MOVE = 24, ///< move down continuously.
    RANDOM = 25, ///< random effects.
    HT_LEFT_SERIES_MOVE = 26, ///< Move left end to end.
    HT_RIGHT_SERIES_MOVE = 27, ///< move right end to end.
    HT_UP_SERIES_MOVE = 28, ///< Move up continuously end to end.
    HT_DOWN_SERIES_MOVE = 29, ///< Move down continuously end to end.
    EFFECT_COUNTS = 30, ///< total number of special effects.
};
```

effect Effect tag

```
<!--
##in Entry special effect ID
##out appearance special effect ID
##inSpeed entry special effect speed { [0-8] 0 is the fastest, 8 is the slowest }
##outSpeed Appearance special effect speed { [0-8] 0 is the fastest, 8 is the slowest }
##duration dwell time { [0-9999] the unit is 0.1 second }
-->
<effect in="##in" out="##out" inSpeed="##inSpeed" outSpeed="##outSpeed"
duration="##duration"/>
```

font Font function node

```
<!--
##name font name (optional)
##italic italic { "true", "false"(default) }
##bold bold { "true", "false" (default) }
##underline underline { "true", "false" (default) }
##size font size
##color font color { "#FF0000" (default red) }
-->
<font name="##name" italic="##italic" bold="##bold" underline="##underline" size="##size"
color="##color"/>
```

style Style type function node

```
<!--
##valign vertical alignment { middle: vertical center; top: top alignment; bottom: bottom
alignment }
##align horizontal alignment { center : center alignment; left: left alignment; right: right
alignment }
-->
<style valign="##valign" align="##align"/>
```

file File node

```
<!--Specify the file to use
##name file name
##md5 file md5 value (optional)
-->
<file name="##name" md5="##md5"/>
```

3. Boot screen function

3.1 BootLogo Start screen

Boot screen related SDK

GetBootLogo Gets the status of the boot screen

```
<!--request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
    <in method="GetBootLogo"/>
</sdk>

<!--Feedback
##exist Whether to set the boot screen {"true": set, "false": not set}
##md5 md5 value of boot image
##name boot image name
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="0a35b47e0821c4ec26d3075b9737a3b5">
    <out method="GetBootLogo" result="kSuccess">
        <logo exist="##exist" md5="##md5" name="##name"/>
    </out>
</sdk>
```

SetBootLogoName Set startup screen

```
<!--request
##md5 md5 value of boot image, can be empty
##name boot image name
-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
    <in method="SetBootLogoName">
        <logo md5="##value" name="##name"/>
    </in>
</sdk>

<!--Feedback-->
```

```
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="0a35b47e0821c4ec26d3075b9737a3b5">
  <out method="SetBootLogoName" result="kSuccess"/>
</sdk>
```

ClearBootLogo clear splash screen

```
<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
  <in method="ClearBootLogo"/>
</sdk>

<!--Feedback-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="0a35b47e0821c4ec26d3075b9737a3b5">
  <out method="ClearBootLogo" result="kSuccess"/>
</sdk>
```

4、Network function

4.1 Get network information

Obtain the SDK related to network information

GetNetworkInfo Obtain network function

```
<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
  <in method="GetNetworkInfo"/>
</sdk>

<!--Feedback
##mode Current network device accessor, priority eth, secondary pppoe, last wifi
##eth.valid and ##pppoe.valid and ##wifi.valie {"true": connected, "false" not connected},
detailed information exists when connected
##Network-related details are detailed in the corresponding modules below
-->
<?xml version="1.0" encoding="utf-8"?>
```



```

<sdk guid="0a35b47e0821c4ec26d3075b9737a3b5">
  <out method="GetNetworkInfo" result="kSuccess">
    <network mode="##mode"/>
    <!--eht access display details -->
    <eth valid="##valid">
      <enable value="##value"/>
      <dhcp auto="##auto"/>
      <address netmask="##netmask" gateway="##gateway" ip="##ip"
dns="##dns"/>
    </eth>
    <!--pppoe is not connected, there is no detailed information-->
    <pppoe valid="##valid"/>
    <!--wifi is not connected, there is no detailed information -->
    <wifi valid="##valid"/>
  </out>
</sdk>

```

GetEth0Info Only get eth0 information

```

<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
  <in method="GetEth0Info"/>
</sdk>

<!--Feedback
##valid Whether the wired network is connected {"true": connected, "false" not connected},
no detailed information when false
##enable.value {"true": connected, "false" not connected}
##dhcp.auto {"true" (dhcp obtains ip address), "false" (static ip address)}
##address.ip ip address
##address.netmask netmask
##address.gateway Gateway
##address.dns dns address
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="0a35b47e0821c4ec26d3075b9737a3b5">
  <out method="GetEth0Info" result="kSuccess">
    <eth valid="##valid">
      <enable value="##value"/>
      <dhcp auto="##auto"/>
      <address ip="##ip" netmask="##netmask" gateway="##gateway"
dns="##dns"/>
    </eth>

```

```
</out>
</sdk>
```

GetPppoeInfo Only get pppoe(3/4G) information

```
<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
  <in method="GetPppoeInfo"/>
</sdk>

<!--Feedback
##valid Whether there is access {"true" (with 3/4G module access), "false" (without 3/4G
module access)}
##enable.value {"true"(with 3/4G network access), "false"(without 3/4G network access)}
##apn.value apn value
##manufacturer.value: Manufacturer of the module
##version.value: module version
##model.value: module model
##imei.value: Module IMEI
##number.value: SIM card phone number
##operators.value: operator
##signal.value: Signal strength, value range [1, 5]; 1 means the worst signal strength; 5
means the best signal strength
##dbm.value: signal strength (in dbm)
##insert.value: Whether the SIM card is inserted, the value range {"true" (with a SIM card
inserted), "false" (without a SIM card inserted)}
##status.value: network registration status, value range {"unregister" (unregistered),
"register local" (registered, local network), "searching" (searching), "reject" (reject
registration), "unknow" (unknown error), "register roaming" (registered, roaming network),
"init" (initialization state)}

##network.value: Network standard, value range {"init" (initialization state), "unknown"
(unknown network), "2G" (2G), "2.5G" (2.5G), "3GPP" (3GPP Family), "3G TD"(Mobile 3G),
"3.5G HSDPA", "3.5G HSUPA", "3.5G HSPAPlus", "4G LTE", "4G TDD", "4G FDD"}

##code.value: error code (reserved)
-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="0a35b47e0821c4ec26d3075b9737a3b5">
  <out method="GetPppoeInfo" result="kSuccess">
    <pppoe valid="##value">
      <enable value="##value"/>
      <apn value="##value"/>
```

```

        <manufacturer value="##value"/>
        <version value="##value"/>
        <model value="##value"/>
        <imei value="##value"/>
        <number value="##value"/>
        <operators value="##value"/>
        <signal value="##value"/>
        <dbm value="##value"/>
        <insert value="##value"/>
        <status value="##value"/>
        <network value="##value"/>
        <code value="##value"/>
    </pppoe>
</out>
</sdk>

```

GetWifiInfo Only get wifi information

```

<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
    <in method="GetWifiInfo"/>
</sdk>

<!--Feedback
##wifi.valid value range {"true" (with WIFI module access), "false" (without WIFI module access)};
##enable.value value range {"true" (with WIFI network access), "false" (without WIFI network access)}
##mode.value: wifi working mode, value range {"ap"(ap mode), "station"(client mode)}
##ap.ssid.value: ssid, fixed value (device ID)
##ap.passwd.value: password
##ap.channel.value: channel
##ap.encryption.value: encryption method, fixed value "WPA-PSK"
##ap.dhcp: ignore
##ap.address: ignore
##station.current.index: indicates the index value of the currently used ap node in the list, and -1 indicates that the ap node is not selected
##station.list contains multiple items, item represents wifi discovery and saved ap node information
##station.list.item.ssid.value: AP's ssid
##station.list.item.passwd.value: password
##station.list.item.signal.value: Signal strength
##station.list.item.apmac.value: The mac address of the ap device

```

```

##station.list.item.dhcp.auto: Value range {"true" (dhcp obtains ip address), "false" (static ip
address)}
##station.list.item.address.ip: WIFI network ip address
##station.list.item.address.netmask: WIFI network subnet mask
##station.list.item.address.gateway: default route
##station.list.item.address.dns: dns server address
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="0a35b47e0821c4ec26d3075b9737a3b5">
  <out method="GetWifiInfo" result="kSuccess">
    <wifi valid="##value">
      <enable value="##value"/>
      <mode value="##value"/>
      <ap>
        <ssid value="##value"/>
        <passwd value="##value"/>
        <channel value="##value"/>
        <encryption value="##value"/>
        <dhcp auto="##value"/>
        <address ip="##value" netmask="##value" gateway="##value"
dns="##value"/>
      </ap>
    </wifi>
    <station>
      <current index="##value"/>
      <list>
        <item>
          <ssid value="##value"/>
          <passwd value="##value"/>
          <signal value="##value"/>
          <apmac value="##value"/>
          <dhcp auto="##value"/>
          <address ip="##value" netmask="##value" gateway="##value"
dns="##value"/>
        </item>
      </list>
    </station>
  </out>
</sdk>

```

4.2

Set network information

Set the SDK related to network information

When modifying network configuration, some will cause network disconnection.

SetEth0Info Set eth0 information

```
<!--request
##valid fixed value "true"
##enable.value fixed value "true"
##dhcp.auto {"true" (dhcp obtains ip address), "false" (static ip address)}
##address.ip ip address
##address.netmask netmask
##address.gateway Gateway
##address.dns dns address -->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
  <in method="SetEth0Info">
    <eth valid="##valid">
      <enable value="##value"/>
      <dhcp auto="##auto"/>
      <address ip="##ip" netmask="##netmask" gateway="##gateway"
dns="##dns"/>
    </eth>
  </in>
</sdk>

<!--Feedback-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="0a35b47e0821c4ec26d3075b9737a3b5">
  <out method="SetEth0Info" result="kSuccess">
  </out>
</sdk>
```

SetWifiInfo Set wifi information

```
<!--request
##mode.value: wifi working mode, value range {"ap"(ap mode), "station"(client mode)}
##ap.ssid.value: ssid, fixed value (device ID)
##ap.passwd.value: password
##ap.channel.value: channel
##ap.encryption.value: encryption method, fixed value "WPA-PSK"
##ap.dhcp.auto: fixed value "true"
```

```

##ap.address: fixed value ip="0.0.0.0" netmask="0.0.0.0" gateway="0.0.0.0" dns="0.0.0.0"
##station.current.index: indicates the index value of the currently used ap node in the list,
and -1 indicates that the ap node is not selected
##station.list contains multiple items, item represents wifi discovery and saved ap node
information
##station.list.item.ssid.value: AP's ssid
##station.list.item.passwd.value: password
##station.list.item.signal.value: Signal strength
##station.list.item.apmac.value: The mac address of the ap device
##station.list.item.dhcp.auto: Value range {"true" (dhcp obtains ip address), "false" (static ip
address)}
##station.list.item.address.ip: WIFI network ip address
##station.list.item.address.netmask: WIFI network subnet mask
##station.list.item.address.gateway: default route
##station.list.item.address.dns: dns server address
-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
  <in method="SetWifiInfo">
    <mode value="##value"/>
    <ap>
      <ssid value="##value"/>
      <passwd value="##value"/>
      <channel value="##value"/>
      <encryption value="##value"/>
      <dhcp auto="##value"/>
      <address gateway="##value" netmask="##value" dns="##value"
ip="##value"/>
    </ap>
    <station>
      <current index="##value"/>
      <list>
        <item>
          <ssid value="##value"/>
          <passwd value="##value"/>
          <signal value="##value"/>
          <apmac value="##value"/>
          <dhcp auto="##value"/>
          <address gateway="##value" netmask="##value" dns="##value"
ip="##value"/>
        </item>
      </list>
    </station>
  </in>

```

```

</sdk>

<!--Feedback-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="0a35b47e0821c4ec26d3075b9737a3b5">
  <out method="SetWifiInfo" result="kSuccess">
  </out>
</sdk>

```

SetApn Set Apn

```

<!--Request
apn.value: apn
-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
  <in method="SetApn">
    <apn value="##value"/>
  </in>
</sdk>

<!--Feedback-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="0a35b47e0821c4ec26d3075b9737a3b5">
  <out method="SetApn" result="kSuccess"/>
</sdk>

```

4.3 Test network

Test network-related SDK

TestAPMode Turn on test AP mode

```

<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
  <in method="TestAPMode"/>
</sdk>

<!--Feedback-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="0a35b47e0821c4ec26d3075b9737a3b5">
  <out method="TestAPMode" result="kSuccess"/>

```

```
</sdk>
```

TestApModeResult Test AP mode result

```
<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
  <in method="TestApModeResult"/>
</sdk>

<!--Feedback
##result When kInvalidParam is returned, the test mode is not enabled
When kNotFoundWifiModule is returned, the wifi module is not found
When kTestWifiUnsuccessful is returned, the AP mode test failed
When kSuccess is returned, it is successful
-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="0a35b47e0821c4ec26d3075b9737a3b5">
  <out method="TestApModeResult" result="##result"/>
</sdk>
```

TestApDisconnect Stop test AP mode

```
<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
  <in method="TestApDisconnect"/>
</sdk>

<!--Feedback-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="0a35b47e0821c4ec26d3075b9737a3b5">
  <out method="TestApDisconnect" result="kSuccess"/>
</sdk>
```

TestStationMode Test Station mode

```
<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
  <in method="TestStationMode"/>
</sdk>
```



```

<!--Feedback-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="0a35b47e0821c4ec26d3075b9737a3b5">
  <out method="TestStationMode" result="kSuccess"/>
</sdk>

```

TestStationModeResult Test Station mode result

```

<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
  <in method="TestStationModeResult"/>
</sdk>

<!--Feedback
##result When kInvalidParam is returned, the test mode is not enabled
When kNotFoundWifiModule is returned, the wifi module is not found
When kTestWifiUnsuccessful is returned, the AP mode test failed
When kSuccess is returned, it is successful
-->
  <?xml version='1.0' encoding='utf-8'?>
<sdk guid="0a35b47e0821c4ec26d3075b9737a3b5">
  <out method="TestStationModeResult" result="##result"/>
</sdk>

```

TestStationDisconnect Stop test Station mode

```

```xml <?xml version='1.0' encoding='utf-8'?>

<?xml version='1.0' encoding='utf-8'?>

```

## 5. Font function

### Get or reload fonts

### GetAllFontInfo Get font information

```

<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="GetAllFontInfo"/>
</sdk>

<!--Feedback
font.name: font name
font.file: font file name
font.bold: Whether the font supports bold, value range {"true" (support), "false" (not
support)}
font.italic: Whether the font supports italics, the value range {"true" (support), "false" (not
support)}
font.underline: Whether the font supports underline, value range {"true" (support), "false"
(not support)}
-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="fecc9420c96a6a708b35e13f62385e15">
 <out method="GetAllFontInfo" result="kSuccess">

 <font bold="##bold" italic="##italic" underline="##underline" name="##name"
file="##file"/>

 </out>
</sdk>

```

## ReloadAllFonts Reload all fonts

```

<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="ReloadAllFonts"/>
</sdk>

<!--Feedback-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="fecc9420c96a6a708b35e13f62385e15">
 <out method="ReloadAllFonts" result="kSuccess"/>
</sdk>

```

## 6. General functions

### Obtain information about the device

## GetDeviceInfo Get device information

```
<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="GetDeviceInfo"/>
</sdk>

<!--Feedback
device.cpu: device cpu type
device.model: device type name
device.id: Device ID
device.name: device name
version.fpga: fpga firmware version number
version.app: firmware version number of the lower computer
version.kernel: Kernel version number
screen.width: screen width
screen.height: screen height
screen.rotation: rotation flag, value range {"0", "90", "180", "270"}
-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="fecc9420c96a6a708b35e13f62385e15">
 <out method="GetDeviceInfo" result="kSuccess">
 <device cpu="##cpu" model="##model" name="##name" id="##id"/>
 <version fpga="##fpga" kernel="##kernel" app="##app"/>
 <screen width="##width" height="##height" rotation="##rotation"/>
 </out>
</sdk>
```

## GetDeviceName Get device name

```
<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="GetDeviceName"/>
</sdk>

<!--Feedback
##value Device name
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="fecc9420c96a6a708b35e13f62385e15">
 <out method="GetDeviceName" result="kSuccess">
 <name value="##value"/>
 </out>
</sdk>
```

```
</out>
</sdk>
```

## GetHardwareInfo Get hardware information

```
<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="GetHardwareInfo"/>
</sdk>

<!--Feedback-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="fecc9420c96a6a708b35e13f62385e15">
 <out method="GetHardwareInfo" result="kSuccess">
 <cpu model="" info=""/>
 <memory model="" size="0" info=""/>
 <storage>
 <flash model="" free="0" total="0" info=""/>
 <extend model="" free="0" total="0" info=""/>
 </storage>
 <eth model="" info=""/>
 <system kernel="" info=""/>
 <ExternInterface>
 <gps model="" valid="false" info=""/>
 <pppoe model="" valid="false" info=""/>
 <wifi model="" valid="false" info=""/>
 <sensor model="" valid="false" info=""/>
 </ExternInterface>
 </out>
</sdk>
```

## 6.2 Set device-related information

### SetDeviceName sets the device name

```
<!--Request
##value Device name
-->
```

```

<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="SetDeviceName">
 <name value="##value"/>
 </in>
</sdk>

<!--Feedback-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="fecc9420c96a6a708b35e13f62385e15">
 <out method="SetDeviceName" result="kSuccess"/>
</sdk>

```

## 6.3 Get or set system volume

### SetSystemVolume sets the system volume

```

<?xml version = "1.0" encoding = "utf-8" ?>
<sdk guid="##GUID">
 <in method="SetSystemVolume" >
 <mode value="##value" />
 <volume percent="##percent" />
 <!-- The node of ploy is optional, and it is not needed when the mode is default -->
 <ploy>
 <item enable="##enable" start="##start" percent="##percent"/>
 </ploy>
 </in>
</sdk>

<!--
##value selection mode default and plays
##percent is the volume value [0-100]
##ploy.item.enable true to enable this configuration item, false to disable
##ploy.item.start time format hh:mm:ss
##ploy.item.percent Volume
-->

<!--Device Answer-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="b040b83a8de5012816e1c2ca1b727305">
 <out method="SetSystemVolume" result="kSuccess"/>
</sdk>

```

## GetSystemVolume Get system volume

```
<?xml version = "1.0" encoding = "utf-8" ?>
<sdk guid="##GUID">
 <in method="GetSystemVolume" />
</sdk>

<!--
device answer
mode tag value is the volume mode
The volume tag percent is the volume value
The play label is an invalid node in default mode
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="b040b83a8de5012816e1c2ca1b727305">
 <out method="GetSystemVolume" result="kSuccess">
 <mode value="default"/>
 <volume percent="100"/>
 <play/>
 </out>
</sdk>
```

## 6.4 Get and set Tcp server

### GetSDKTcpServer Get Tcp server information

```
<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="GetSDKTcpServer"/>
</sdk>

<!--Feedback
##port port
##host host address
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="dedef080ba2082110c73404cceb17135">
 <out result="kSuccess" method="GetSDKTcpServer">
 <server port="##port" host="##host"/>
 </out>
```

```
</sdk>
```

## SetSDKTcpServer Set Tcp server

It is used for the function that requires the device to connect to the tcp server instead of actively connecting to the device

```
<!--request
##port port
##host host address
-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="SetSDKTcpServer">
 <server port="##port" host="##host"/>
 </in>
</sdk>

<!--Feedback-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="dedef080ba2082110c73404cceb17135">
 <out result="kSuccess" method="SetSDKTcpServer"/>
</sdk>
```

## 6.5 Restart device

```
<!--Request
##delay Restart serval seconds
-->
<?xml version = "1.0" encoding = "utf-8" ?>
<sdk guid="##GUID">
 <in method="Reboot" delay="##delay" />
</sdk>

<!--Feedback-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="3b03a3534c3a37c76bd610fee8bc8055">
 <out method="Reboot" result="kSuccess"/>
</sdk>
```

## 6.6 About configuring Socket information

No modification is required under normal circumstances. Try not to modify it to avoid causing the Socket to be continuously closed by the device

### GetSocketTimeInfo Get Socket information

```
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="GetSocketTimeInfo"/>
</sdk>
```

### SetSocketTimeInfo Set Socket information

```
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="SetSocketTimeInfo">
 <heartbeat timeout="##timeout" delay="##delay"/>
 <connctet delay="##delay"/>
 </in>
</sdk>
```

## 7

## 8、 license information

### GetLicense Get license

```
<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="GetLicense"/>
</sdk>

<!--Feedback
-->
<?xml version="1.0" encoding="utf-8"?>
```



```

<sdk guid="f96957643084953befcbd2c1a16f7617">
 <out method="GetLicense" result="kSuccess">
 <time value="0"/>
 <code value=""/>
 <old value=""/>
 <set value="0"/>
 </out>
</sdk>

```

## Set License (Redundant)

SetLicense method, no effect

## Empty License (Redundant)

ClearLicense method, no effect

# 9. Brightness setting

## GetLuminancePloy Get the brightness configuration

```

<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="GetLuminancePloy"/>
</sdk>

```

```

<!--Feedback
##mode.value "default" (default), "ploys" (custom mode), "sensor" (sensor mode)
default mode

```

```

##default.value brightness value [1, 100]
sensor mode
##sensor.min minimum brightness level
##sensor.max maximum brightness level
##sensor.time Brightness adjustment interval time
custom mode
##play can contain multiple item child nodes
##item.enable Whether the item is enabled, {"true" (enabled), "false" (disabled)}
##item.start Change the start time of the brightness level, format: hh:mm:ss
##item.percent: brightness level
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="f96957643084953befcbd2c1a16f7617">
 <out method="GetLuminancePloy" result="kSuccess">
 <mode value="default"/>
 <default value="100"/>
 <sensor min="1" max="100" time="5"/>
 <ploy>
 <item enable="##enable" start="##start" percent="##percent"/>
 </ploy>
 </out>
</sdk>

```

## SetLuminancePloy Set brightness profile

```

<!--Request
##mode.value range {"default" (default), "ploys" (custom mode), "sensor" (sensor mode)}
default mode
##default.value: In the default mode, set the brightness level [1, 100]
custom mode
##play can contain multiple item sub-nodes, indicating that different brightness levels can
be set in different time periods
##item.enable: Whether the item is enabled, value range {"true" (enabled), "false"
(disabled)}
##item.start: The start time of changing the brightness level, format: hh:mm:ss
##item.percent: brightness level
sensor mode
##sensor.min: sensor mode minimum brightness level
##sensor.max: sensor mode maximum brightness level
##sensor.time: Interval time of brightness adjustment, value range [5, 15] (unit second)
-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="SetLuminancePloy">

```

```

 <mode value="##value"/>
 <default value="##value"/>
 <ploy>
 <item enable="##enable" percent="##percent" start="##start"/>
 </ploy>
 <sensor max="##max" min="##min" time="##time"/>
</in>
</sdk>

<!--Feedback-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <out method="SetLuminancePloy" result="kSuccess"/>
</sdk>

```

# 10. Functions related to media files

## 1、DeleteFiles Delete files from device

Compatible with old versions, increase md5 value, delete specified files accurately

Do not specify the md5 value, the same as the old version, but when encountering the same name file, delete the first one

```

<!-- Old version, delete the first file with the same name -->
<?xml version="1.0"?>
<sdk guid="##GUID">
 <in method="DeleteFiles">
 <files>
 <file name="##name"/>
 </files>
 </in>
</sdk>

```

```

<!-- Newly added attribute, accurately delete the specified file -->
<?xml version="1.0"?>
<sdk guid="##GUID">
 <in method="DeleteFiles">
 <files>
 <file name="##name" md5="##md5"/>
 </files>
 </in>
</sdk>

<!--Both Feedback
If the deletion is successful, there will be no files label and sub-label. If the deleted file is not
found, the files label will be returned, and the sub-label will be the file not found.
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="d77658a8565669346a60862dba08f2e1">
 <out result="kSuccess" method="DeleteFiles">
 <files>
 <file result="kFileNotFound" name="##name"/>
 </files>
 </out>
</sdk>

```

## Delete files according to md5

Use the same SDK as DeleteFiles, this is a new feature, instead of using the name attribute, use md5 to delete files.

```

<!-- Only use md5 to delete files -->
<?xml version="1.0"?>
<sdk guid="##GUID">
 <in method="DeleteFiles">
 <files>
 <file md5="##md5"/>
 </files>
 </in>
</sdk>

<!--Feedback
If the deletion is successful, there will be no files label and sub-label. If the deleted file is not
found, the files label will be returned, and the sub-label will be the file not found.
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="d77658a8565669346a60862dba08f2e1">
 <out result="kSuccess" method="DeleteFiles">

```

```

 <files>
 <file result="kFileNotFound" md5="##md5"/>
 </files>
 </out>
</sdk>

```

## 10.2

# AddFiles adds files to the device

The device supports http download

```

<!--request
##type image and video
##size file size
##md5 md5 value of the file
##name file name
##remote resource location, http://... https://...
-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="AddFiles">
 <files>
 <file type="##type" size="##size" md5="##md5" name="##name"
remote="remote"/>
 </files>
 </in>
</sdk>

<!--Feedback-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="41f8d777a9d83984e49fce04ea4c77d0">
 <out method="AddFiles" result="kSuccess">
 <files/>
 </out>
</sdk>

```

## 10.3

# GetFiles GetFiles Get device media file information

```

<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="GetFiles"/>
</sdk>

<!--Feedback
##existSize The size of the file in the lower computer
##md5 md5 value of the file
##name file name
##size file size
##type type
0-image 1-video 2-font 3-firmware 4-fpga configuration file
5-Setting Configuration File 6-Project Resources 7-Data 8-Temporary Files
128-Temporary picture file 129-Temporary video file
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="41f8d777a9d83984e49fce04ea4c77d0">
 <out method="GetFiles" result="kSuccess">
 <files>
 <file existSize="##existSize" type="##type" md5="##md5" name="##name"
size="##size"/>
 </files>
 </out>
</sdk>

```

# 11 Screen switch function

## 11.1 Configure switch screen speed

### GetSpeedForOFFOrON Get SDK switch screen speed

```
<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="GetSpeedForOFFOrON"/>
</sdk>

<!--Feedback
##enable.value SDK switch screen speed limit enable state
##off.speed The speed when the SDK turns off the screen (unit: km/h)
##on.speed The speed when the SDK opens the screen (unit: km/h)
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="bd68900da93b59cc553026851cc05afb">
```

```

 <out result="kSuccess" method="GetSpeedForOFFOrON">
 <enable value="##value"/>
 <off speed="##speed" />
 <on speed="##speed" />
 </out>
 </sdk>

```

## SetSpeedForOFFOrON Set the SDK switch screen speed

```

<!--request
##enable.value SDK switch screen speed limit enable state
##off.speed The speed when the SDK turns off the screen (unit: km/h)
##on.speed The speed when the SDK opens the screen (unit: km/h)
-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="SetSpeedForOFFOrON">
 <enable value="##value"/>
 <off speed="##speed" />
 <on speed="##speed" />
 </in>
</sdk>

<!--Feedback-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="bd68900da93b59cc553026851cc05afb">
 <out result="kSuccess" method="SetSpeedForOFFOrON"/>
</sdk>

```



## 12.Others function

### 12.1GPS Information reporting

#### GetGpsRespondEnable Obtain GPS information reporting enable flag

```
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="GetGpsRespondEnable"/>
</sdk>

<!--Feedback
##enable Whether to enable {"true", "false"}
##delay GPS information upload interval (unit: second)
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="7a8fe8bb88ccfab7dc57adc9c5f70e2e">
 <out result="kSuccess" method="GetGpsRespondEnable">
 <gps enable="##enable" delay="##delay"/>
 </out>
</sdk>
```

```
</out>
</sdk>
```

## SetGpsRespondEnable Set GPS information reporting enable flag

```
<!--Request
##enable Whether to enable {"true", "false"}
##delay GPS information upload interval (unit: second)
-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="SetGpsRespondEnable">
 <gps enable="##enable" delay="##delay"/>
 </in>
</sdk>

<!--Feedback-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="7a8fe8bb88ccfab7dc57adc9c5f70e2e">
 <out result="kSuccess" method="SetGpsRespondEnable"/>
</sdk>
```

## GPS information message

When the secondary development user calls the "SetGpsRespondEnable" interface to enable the GPS information reporting enable flag, when the control card establishes a TCP connection with the server, it will actively upload the GPS information report to the server periodically according to the interval set by the secondary development user. arts.

Filed	Length(Len)	Command(Cmd)	Latitude	Longitude	Time	Reserve
Bytes	2 bytes	2 bytes	4 bytes	4 bytes	4bytes	4 bytes
Values	20	kGPSInfoAnswer	latitude	longitude	seconds	reserved

```
typedef struct HGPSInfoAnswer
{
 uint16 len; ///
```

```

 hfloat longitude; ///< longitude
 uint32 seconds; ///

```

## 12.2 Multi-screen sync flag

### GetMulScreenSync Get Multi-screen sync flag

```

<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="GetMulScreenSync"/>
</sdk>

<!--Feedback
##value Multi-screen synchronization enable flag, value range {"true", "false"}
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="7a8fe8bb88ccfab7dc57adc9c5f70e2e">
 <out result="kSuccess" method="GetMulScreenSync">
 <enable value="##value"/>
 </out>
</sdk>

```

## SetMulScreenSync Set multi-screen synchronization flag

```
<!--request
##value Multi-screen synchronization enable flag, value range {"true", "false"}
-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="SetMulScreenSync">
 <enable value="##value"/>
 </in>
</sdk>

<!--request
##value Multi-screen synchronization enable flag, value range {"true", "false"}
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="7a8fe8bb88ccfab7dc57adc9c5f70e2e">
 <out result="kSuccess" method="SetMulScreenSync" />
</sdk>
```

## 12.3 Program play counts flag

### GetPlayProgramCountsEnable Get Program play counts flag

```
<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="GetPlayProgramCountsEnable"/>
</sdk>

<!--Feedback
##value program playback statistics flag, value range {"true", "false"}
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="7a8fe8bb88ccfab7dc57adc9c5f70e2e">
 <out result="kSuccess" method="GetPlayProgramCountsEnable">
```

```

 <enable value="##value"/>
 </out>
</sdk>

```

## SetPlayProgramCountsEnable Set Program play counts flag

```

<!--Request
##value Program playback statistics flag, value range {"true", "false"}
-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="SetPlayProgramCountsEnable">
 <enable value="##value"/>
 </in>
</sdk>

<!--Feedback
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="7a8fe8bb88ccfab7dc57adc9c5f70e2e">
 <out result="kSuccess" method="SetPlayProgramCountsEnable"/>
</sdk>

```

## 12.4 Program playback statistics file name

### GetPlayProgramCountsFileName Get program playback statistics file name

```

<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="GetPlayProgramCountsFileName"/>
</sdk>

<!--Feedback
##file.name: Program playback statistics file name
-->

```

```
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="7a8fe8bb88ccfab7dc57adc9c5f70e2e">
 <out result="kSuccess" method="GetPlayProgramCountsFileName">
 <file name="##name"/>
 </out>
</sdk>
```

# 13

## 13.1 Program function

1. For program composition, please go to Program Definition Structure to view

## Program Edit

### AddProgram Add program

For program-related nodes, please go to Program Structure Definition to view

guid - Globally unique identifier. Two identical guides are not allowed for the same show

```
<!--Request-->
```

```

<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="AddProgram">
 <!--The following is the program list information, please go to the program
structure definition for details-->
 <screen timeStamps="0">
 <program guid="##program-guid" type="normal">
 <backgroundMusic/>
 <playControl count="1"/>
 <border effect="rotate" speed="4" index="0"/>
 <area alpha="255" guid="##area-guid">
 <rectangle x="0" height="128" width="128" y="0"/>
 <resources>
 <text guid="##text-guid" singleLine="false">
 <style valign="middle" align="center"/>
 <string>example</string>
 <font name="SimSun" italic="false" bold="false"
underline="false" size="28" color="ffffff"/>
 </text>
 </resources>
 </area>
 </program>
 </screen>
 <!--End of program single node here-->
 </in>
</sdk>

<!--Feedback-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="fafaed9fdb9668a5bb06c01f55e95518">
 <out result="kSuccess" method="AddProgram"/>
</sdk>

```

## UpdateProgram Update the corresponding program

```

<!--request
The update program is updated according to the existing program matching GUID.
Make sure that the corresponding guids of ##program-guid, ##area-guid, ##text-guid are
consistent
-->
<?xml version='1.0' encoding='utf-8'?>

```

```

<sdk guid="##GUID">
 <in method="UpdateProgram">
 <!--The following is the program list information, please go to the program structure
definition for details-->
 <program guid="##program-guid" type="normal">
 <playControl count="1"/>
 <border index="0" effect="rotate" speed="4"/>
 <area guid="##area-guid" alpha="255">
 <rectangle x="0" y="0" height="128" width="128"/>
 <resources>
 <text guid="##text-guid" singleLine="false">
 <style align="center" valign="middle"/>
 <string>example</string>
 <font bold="false" name="SimSun" size="28" color="#ffffff"
italic="false" underline="false"/>
 </text>
 </resources>
 </area>
 </program>
 <!--End of program single node here-->
 </in>
 </sdk>

<!--Feedback-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="fafaed9fdb9668a5bb06c01f55e95518">
 <out result="kSuccess" method="UpdateProgram"/>
</sdk>

```

## DeleteProgram Delete program

```

<!--request
Delete programs are deleted according to the matching GUID of existing programs.
Make sure that the corresponding guids of ##program-guid, ##area-guid, ##text-guid are
consistent
-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="DeleteProgram">
 <program guid="##program-guid" type="normal">
 <area guid="##area-guid" alpha="255">
 <resource>
 <text guid="##text-guid" singleLine="false">

```



```

 <string>delete program</string>
 </text>
</resource>
</area>
</program>
</in>
</sdk>

<!--Feedback-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="fafaed9fdb9668a5bb06c01f55e95518">
 <out result="kSuccess" method="DeleteProgram"/>
</sdk>

```

## 13.2RealTimeUpdateRegional real-time update SDK

Currently, only text and image tags are supported for real-time regional updates

The rk series does not support real-time update of special effects.

### Add show demos that require real-time updates of regions

Bind the program that needs to be modified to the guid attribute. And give a unique guid value. In order not to affect viewing, use (...) to omit irrelevant attributes. The demonstration is as follows:

1. Add programs first
2. Add guid to the program that needs to be updated in real time, you can see that the text tag has added the guid attribute. This will be used for real-time update later

image: When the dynamic map and the static map are switched, the data will be replaced in the area, which will cause a flash (the specific length depends on factors such as size and loading speed), but the same nature will not be switched (the dynamic map will be stuck).

```
<?xml version = "1.0" encoding = "utf-8" ?>
```

```

<sdk guid = "##GUID">
 <in method="AddProgram">
 <screen ...>
 <program ...>
 <playControl .../>
 <area ...>
 <rectangle .../>
 <resources>
 <text guid="c544d34e-fd6c-11ea-8edc-00005f6af662" name=""
singleLine="false">
 <string> Text area plus guid attribute </string>
 </text>
 </resources>
 </area>
 </program>
 </screen>
 </in>
</sdk>

```

## RealTimeUpdate Real-time regional updates

The area update requires that the program corresponding to the **guid** must exist, and the program type must be the same type, otherwise an error code will be returned

Allow updating multiple nodes

Supports remote resources, but please note that if there are remote resources that need to be downloaded, the data of the local resources will be updated first, and the remote resources will enter the downloading state, and the result attribute value will be **kDownloadingFile**. If the resource download fails, it will be **kDownloadFileFailed**. Will give back **kSuccess**.

If it belongs to a remote resource, then the file tag must fill in the md5 attribute of the remote resource, otherwise it will feedback **kSeekFileFailed**

Note: When it is in the download state, sending another real-time update that requires resource download will cause the download to fail. And this time the task that requires resource download will be cancelled.

text: requires the same guid to be updated

image: requires the file tag to require the name and the md5 attribute of the image, allowing only the name attribute to be used, but only updating the first one when encountering the same name

```
<!--
The child node of content is the program that needs to be updated
##guid attribute corresponds to the guid of the program that needs to be updated, the two
must be the same, if it does not exist, an error code will be returned
##type attributes are local and remote, local means that the resource is on the local, remote
means that the resource needs to be downloaded, at this time the file tag must have name
and md5 attributes
-->
<?xml version = "1.0" encoding = "utf-8" ?>
<sdk guid = "##GUID">
 <in method="RealTimeUpdate">
 <content guid="c544d34e-fd6c-11ea-8edc-00005f6af662" type="##type" >
 <text guid="c544d34e-fd6c-11ea-8edc-00005f6af662" singleLine="false">
 <string>Hello World</string>

 </text>
 </content>

 <content guid="6e55b916-a9c1-4522-bbfd-63e6c6a63436" type="local">
 <image guid="6e55b916-a9c1-4522-bbfd-63e6c6a63436" fit="stretch">
 <!-- This means that the resource is local -->
 <file name="test.jpg" md5="449fa54b8ad97ac56dd0f4b1d1f66dbd"/>
 </image>
 </content>

 <content guid="6e55b916-a9c1-4522-bbfd-63e6c6a63436" type="remote">
 <image guid="6e55b916-a9c1-4522-bbfd-63e6c6a63436" fit="stretch">
 <!-- This indicates that the resource needs to be downloaded. The type
attribute of content is remote. Because the resource needs to be downloaded, the file tag
needs to be complete at this time, and the md5 attribute must exist -->
 <file name="http://127.0.0.1/test.jpg" md5="449fa54b8ad97ac56dd0f4b1d1f66dbd"/>
 </image>
 </content>

 <!-- If there are still needs to be updated, you can continue to add content tags -->
 <content guid="..." type="##type">
 ...
 </content>
</in>
</sdk>
```

```

<!--Feedback-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="09e83c4a6fca57b429482ccb9e93bd6a">
 <out method="RealTimeUpdate" result="kSuccess"/>
</sdk>

```

## 13.5 ScreenRotation Screen rotation

When setting screen rotation, it will cause width and height adjustment and program replay, please do not rotate when multi-screen is synchronized.

The value value of the rotation tag corresponds to clockwise rotationrotation

0 -> 0°

1 -> 90°

2 -> 180°

3 -> 270°

```

<!--
screen rotation request
##value [0-3], returns kInvalidXmlIndex if it does not belong
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="##GUID">
 <in method="ScreenRotation">
 <rotation value="##value"/>
 </in>
</sdk>

<!--
Screen rotation feedback-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="d77e142ea74b179d648e497ea820e6e2">
 <out result="kSuccess" method="ScreenRotation"/>
</sdk>

```

13.6

## SwitchProgram Program switching

It can use the guid of the program or use the index to switch the program

```

<!--

```

Program switching request, just choose one attribute for guid and index  
For example, when there is only one guid attribute and index attribute  
##guid If the guid does not exist in the program, kInvalidXmlIndex will be returned  
##index [0-max) program index, return kInvalidXmlIndex when max is exceeded  
If both exist, the guid attribute has a higher priority than the index, and the index will be used only when the guid cannot be found

```
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="##GUID">
 <in method="SwitchProgram">
 <!-- choose one of the guid attribute and index attribute -->
 <program guid="##guid" index="##index"/>
 </in>
</sdk>

<!--
Program switching feedback
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="d77e142ea74b179d648e497ea820e6e2">
 <out result="kSuccess" method="SwitchProgram"/>
</sdk>
```

## 13.6 Get the GUID of the currently playing program

```
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="GetCurrentPlayProgramGUID"/>
</sdk>

<!--
##program.guid The guid of the currently playing program
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="fafaed9fdb9668a5bb06c01f55e95518">
 <out result="kSuccess" method="GetCurrentPlayProgramGUID">
 <program guid="##guid"/>
 </out>
</sdk>
```

## **14 Screen function**

### **14.1 GetScreenshot and GetScreenshot2 Screenshots**

There are two ways to take a screenshot of the current screen. One is to poll the device to see if the screenshot is complete, and the second is to asynchronously wait for the device to reply to the screenshot. It is recommended to use the second

#### **GetScreenshot Screenshot of polling operation**

Operating procedures:

1. Use GetScreenshot to notify the lower computer to start taking screenshots
2. Use GetScreenshotStatus to get whether the image data exists
3. Use GetScreenshotData to get image data

```
<!--
 1. Clean up the screenshot data, and notify the lower computer to start capturing a
 screenshot
 2. ##width value range {-int, int}, default (less than or equal to 0, adaptive, larger than
 frame buffer range, adaptive)
 3. ##height value range {-int, int}, default (less than or equal to 0, adaptive, larger than
 frame buffer range, adaptive)
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="##GUID">
 <in method="GetScreenshot">
 <image width="##width" height="##height"/>
 </in>
</sdk>

<!-- parameter response
 1. ##result kCreateFileFailed means that the screen capture fails, usually because
 the program is not playing
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="##GUID">
 <out method="GetScreenshot" result="kSuccess">
 </out>
</sdk>
```

## Get the screenshot status of the lower computer, which is used to assist GetScreenshot

```
<!--
 1. Obtain the screenshot status of the lower computer, and judge whether there is
 screenshot data in the lower computer
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="##GUID">
```

```

 <in method="GetScreenshotStatus">
 </in>
</sdk>

<!-- parameter response
 1. If ##result is kProcessing, there is no picture data. If GetScreenshot is sent, it is
 taking a screenshot, and kSuccess has screenshot data
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="##GUID">
 <out method="GetScreenshotStatus" result="kSuccess">
 </out>
</sdk>

```

## Obtain the screenshot data of the lower computer, which is used to assist GetScreenshot

```

<!--
 1. Obtain the screenshot data of the lower computer
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="##GUID">
 <in method="GetScreenshotData">
 </in>
</sdk>

<!-- parameter response
 1. ##data data is the data converted from pictures to base64, and the restored data
 content is jpg picture data
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="##GUID">
 <out method="GetScreenshotData" result="kSuccess">
 <image data="##data"/>
 </out>
</sdk>

```



# GetScreenshot2 Screenshot of asynchronous operation

GetScreenshot2 is asynchronous screenshot, when you request, you can continue to do other things, it will not block the device operation.

At the same time, this is a different cache from GetScreenshot, which means you can't use the above to check whether the device has screenshot data, and there is no need to do this

After the device completes the screenshot operation, it will automatically feed back the data to you without requesting it again. Of course, this feedback speed is the sum of the reply time and the time spent on capturing the screen.

```
<!--
 1. Clean up the screenshot data, and take a screenshot from the lower computer. After
the screenshot is completed, it will reply the data
 2. ##width value range {-int, int}, default (less than or equal to 0, adaptive, larger than
frame buffer range, adaptive)
 3. ##height value range {-int, int}, default (less than or equal to 0, adaptive, larger than
frame buffer range, adaptive)
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="##GUID">
 <in method="GetScreenshot2">
 <image width="##width" height="##height"/>
 </in>
</sdk>

<!-- parameter response
 1. ##data data is the data converted from pictures to base64, and the restored data
content is jpg picture data
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="##GUID">
 <out method="GetScreenshot2" result="kSuccess">
 <image data="##data"/>
 </out>
</sdk>
```

# ScreenShot (deprecated)

screen

program

Only supported by some devices

```
<!--Request-->
<?xml version = "1.0" encoding = "utf-8" ?>
<sdk guid="##GUID">
 <in method="ScreenShot" />
</sdk>

<!--Feedback
##image base64 image data
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="624aedc8936411b50636e01248789d98">
 <out method="ScreenShot" result="kSuccess">
 <screenshot image="##image"/>
 </out>
</sdk>
```

## 15.Screen switch function

### 15.1 Switch screen information

**GetSwitchTime:** Get Switch screen information

```

<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="GetSwitchTime"/>
</sdk>

<!--Feedback
##open.enable current screen state
##play.enable Turn on the time period control switch screen {"true" (open), "false" (not
open)}
##item.enable time period whether the item is enabled
##item.start Screen opening time, format hh:mm:ss
##item.end screen off time, format hh:mm:ss
The screen is on during the time period from item.start to item.end
--><?xml version="1.0" encoding="utf-8"?>
<sdk guid="bd68900da93b59cc553026851cc05afb">
 <out result="kSuccess" method="GetSwitchTime">
 <open enable="##enable"/>
 <play enable="##enable">
 <item enable="##enable" end="##end" start="##start"/>
 </play>
 </out>
</sdk>

```

## SetSwitchTime Get Switch screen information

```

<!--request
##open.enable fixed value "true"
##play.enable: Whether to enable the time period control switch screen, the value range
{"true" (enabled), "false" (not enabled)}
##item.enable: Whether the item is enabled, value range {"true" (enabled), "false"
(disabled)}
##item.start: Screen opening time, format hh:mm:ss
##item.end: screen off time, format hh:mm:ss
The screen is on during the time period from item.start to item.end
-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="SetSwitchTime">
 <open enable="##enable"/>
 <play enable="##enable">
 <item enable="##enable" end="##end" start="##start"/>
 </play>
 </in>
</sdk>

```

```

 </in>
 </sdk>

 <!--Feedback-->
 <?xml version='1.0' encoding='utf-8'?>
 <sdk guid="bd68900da93b59cc553026851cc05afb">
 <out result="kSuccess" method="SetSwitchTime"/>
 </sdk>

```

## 15.2 Set switch screen

### OpenScreen :open screen

```

 <!--Request-->
 <?xml version='1.0' encoding='utf-8'?>
 <sdk guid="##GUID">
 <in method="OpenScreen"/>
 </sdk>

 <!--Feedback-->
 <?xml version="1.0" encoding="utf-8"?>
 <sdk guid="09bf0cceca82bb7c8cb494ddb136a25">
 <out result="kSuccess" method="OpenScreen"/>
 </sdk>

```

### CloseScreen :close screen

```

 <!--Close screen-->
 <?xml version='1.0' encoding='utf-8'?>
 <sdk guid="##GUID">
 <in method="CloseScreen"/>
 </sdk>

 <!--Feedback-->
 <?xml version="1.0" encoding="utf-8"?>
 <sdk guid="09bf0cceca82bb7c8cb494ddb136a25">
 <out result="kSuccess" method="CloseScreen"/>
 </sdk>

```

## **16.Sensor related function**

### **16.1 Serial port transparent transmission setting**

1. The Sensor port supports Uart, RS232 and 485
2. RS232 and 485 need adapter board

3. The Sensor port is a Uart interface sensor

## Set485Param Set serial port setting

```
<!--
1. Set the 485 parameter request, if it is closed, all can be set to empty
2. It only needs to be set once, and the parameters of the setting items will be saved
3. The value of ##name is: "sensor" (indicating that the 485 device is connected to the
sensor port); "gps" (indicating that the 485 device is connected to the gps port)
4. ##baudRate Baud rate, the same as the corresponding serial device, generally 9600
5. ##dataBits data bits, the same as the corresponding serial device, generally 8
6. ##stopBits Stop bit, same as the corresponding serial device, generally 1
7. ##parity parity, the same as the corresponding serial device, generally n, means no
parity
8. ##timeout Use the timeout time of ReadDataFrom485, the unit is second (s)
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="##GUID">
 <in method="Set485Param">
 <device name="##name" baudRate="##baudRate" dataBits="##dataBits"
stopBits="##stopBits" parity="##parity"/>
 <read timeout="##timeout"/>
 </in>
</sdk>

<!-- Set 485 parameter response -->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="##guid">
 <out method="Set485Param" result="##result"/>
</sdk>
```

## Get485Param Get serial port settings

```
<!-- Set the 485 parameter request,-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="##GUID">
 <in method="Get485Param"/>
</sdk>

<!-- Set 485 parameter response -->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="##guid">
 <out method="Get485Param" result="##result">
```

```

 <device name="##name" baudRate="##baudRate" dataBits="##dataBits"
stopBits="##stopBits" parity="##parity"/>
 <read timeout="##timeout"/>
</out>
</sdk>

```

## SendDataTo485 Send data to serial device

Added configuration verification, when there is an empty configuration in the 485 serial port configuration and the timeout time  $\leq 0$ , kInvalidParam will be returned

1. Sending data needs to be converted to base64. After sending, the device will convert the original data and forward it to the serial device. It supports both binary data
2. Base64 data replaces the ##data tag below<!--

```

1. Send data to 485 device request
2. ##data is base64 encoding of 485 data
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="##GUID">
 <in method="SendDataTo485">
 <data value="##data"/>
 </in>
</sdk>

<!-- Set 485 parameter response -->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="##guid">
 <out method="SendDataTo485" result="##result"/>
</sdk>

```

## ReadDataFrom485 Read serial device data

Added configuration verification, when there is an empty configuration in the 485 serial port configuration and the timeout time  $\leq 0$ , kInvalidParam will be returned

Note: The device has a 1k buffer, and the previous data will be lost if more than 1k of cached data has not been read. The buffered data always maintains a size of 1k. The buffered data will be cleared until it is read.

```
<!-- Set 485 parameter request -->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="##GUID">
 <in method="ReadDataFrom485"/>
</sdk>

<!--
1. Send data to 485 device request
2. ##data is base64 encoding of 485 data
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="##guid">
 <out method="ReadDataFrom485" result="##result">
 <data value="##data"/>
 </out>
</sdk>
```

## 16.2

# Get sensor information function

## GetSensorInfo Get sensor information function

```
<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="GetSensorInfo"/>
</sdk>

<!--Feedback
##connect connection status value false and true
##luminance Luminance sensor
##temp1 temperature sensor
##humidity Humidity Sensor
##temp2 Temperature 2 sensor
##telecontroller remote control
gps gps
```



```

##windSpeed wind speed
##windDirection wind direction
##noise noise
##pressure air pressure
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="09bf0cceca82bb7c8cb494ddbb136a25">
 <out result="kSuccess" method="GetSensorInfo">
 <luminance connect="##connect"/>
 <temp1 connect="##connect"/>
 <humidity connect="##connect"/>
 <temp2 connect="##connect"/>
 <telecontroller connect="##connect"/>
 <gps connect="##connect"/>
 <windSpeed connect="##connect"/>
 <windDirection connect="##connect"/>
 <noise connect="##connect"/>
 <pressure connect="##connect"/>
 </out>
</sdk>

```

## GetGPSInfo Get GPS information

```

<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="GetGPSInfo"/>
</sdk>

<!--Feedback
##east East {"false", "true"}
##north North {"false", "true"}
##longitude longitude
##latitude latitude
##counts count
##speed speed
##direction direction-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="ae26d72998f2e97eb38dcc7e96a15656">
 <out method="GetGPSInfo" result="kSuccess">
 <gps east="##east" north="##north" longitude="##longitude" latitude="##latitude"
counts="##counts" speed="##speed" direction="##direction"/>
 </out>
</sdk>

```

## GetCurrentSensorValue      Get      current sensor value

```
<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="GetCurrentSensorValue"/>
</sdk>

<!--Feedback
##luminance.current Luminance value
##temperature.current temperature value
##humidity.current Humidity value
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="0f55d95d39606d0162e03efece728a5d">
 <out method="GetCurrentSensorValue" result="kSuccess">
 <luminance current="##current"/>
 <temperature current="##current"/>
 <humidity current="##current"/>
 </out>
</sdk>
```

## 16.3 Serial port SDK

### GetSerialSDK      Get      serial      port      SDK configuration

```
<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="GetSerialSDK"/>
</sdk>

<!--Feedback
##name serial interface name {"sensor", "gps"}
##enable enable serial port SDK function {"true", "false"}
-->
<?xml version="1.0" encoding="utf-8"?>
```

```

<sdk guid="0dda9f5d2f5243075256923b9a8dcde5">
 <out result="kSuccess" method="GetSerialSDK">
 <config name="##name" enable="##enable"/>
 </out>
</sdk>

```

## SetSerialSDK Set the serial port SDK configuration

```

<!--Request
##name serial interface name {"sensor", "gps"}
##enable enable serial port SDK function {"true", "false"}
-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="SetSerialSDK">
 <config name="##name" enable="##enable"/>
 </in>
</sdk>

<!--反馈-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="0dda9f5d2f5243075256923b9a8dcde5">
 <out result="kSuccess" method="SetSerialSDK"/>
</sdk>

```

## Serial port SDK send and receive format

Field	Command packet length(len)	Command(cmd)	Xml total length(total)	Xml current place(index)	xml crc32	xml
Bytes	2 bytes	2 bytes	4 bytes	4 bytes	4 bytes	N bytes

Each package of serial port SDK data is required to be no more than 9000 bytes, and more than 9000 bytes will be divided into packages.

```

#pragma pack(1)
typedef struct HHeader
{
 uint16 len; ///< Command packet length
 uint16 cmd; ///< Command value

```

```

} HHeader;

typedef struct HSerialSDKAsk
{
 HHeader header;
 uint32 total; ///< xml total length
 uint32 index; ///< The position of xml in the total xml, the first package is 0, and the
 ///< next package is the position of cut package>
 uint32 crc32; ///< The crc32 check value of the xml data of the current package
 uint8 data[1];
} HSerialSDKAsk, HSerialSDKAnswer;
#pragma pack()

```

## 17. Time function

### Time correction function

#### GetTimeInfo Get time correction data

```

<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="GetTimeInfo"/>
</sdk>

<!--Feedback
##timezone.value: time zone string, must start with (UTC+hh:mm), for example:
(UTC+08:00)Beijing,Chongqing,HongKong,Urumchi

```

```

##summer.enable: Whether to enable summer time, value range {"true" (enable summer
time), "false" (disable daylight saving time)}
##sync.value: Whether to enable automatic time synchronization, value range {"none"
(automatic synchronization not enabled), "gps_rs232", "gps_usb", "gps_pcie" (gps
calibration time), "network" (network calibration time), "auto" (automatic time adjustment),
"rf" (rf time adjustment)}
Automatic time calibration: Automatically choose to use GPS time calibration when there is
GPS access, otherwise use network time calibration
##time.value: Current control card time, format: "yyyy-mm-dd hh:MM:ss", for example:
"2017-01-01 00:00:00"
##server.list server list, comma separated
##rf.enable.value Whether to enable rf synchronization {"false" disable "true" enable}
##rf.master.value Master {"false", "true"}
##rf.channel.value channel
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="0f55d95d39606d0162e03efece728a5d">
 <out method="GetTimeInfo" result="kSuccess">
 <timezone value="##value"/>
 <summer enable="##enable"/>
 <sync value="##value"/>
 <time value="##time"/>
 <server list="##list"/>
 <rf>
 <enable value="##value"/>
 <master value="##value"/>
 <channel value="##value"/>
 </rf>
 </out>
</sdk>

```

## SetTimeInfo Set time correction data

```

<!--Request
##timezone.value: time zone string, must start with (UTC+hh:mm), for example:
(UTC+08:00)Beijing,Chongqing,HongKong,Urumchi
##summer.enable: Whether to enable summer time, value range {"true" (enable summer
time), "false" (disable daylight saving time)}
##sync.value: Whether to enable automatic time synchronization, value range {"none" (do
not enable automatic synchronization), "gps" (gps calibration), "network" (network
calibration), "auto" (automatic calibration hour)}
##time.value: The time of the device will be set. This value will take effect when
sync.value="none", and will be ignored in other cases. The format is yyyy-MM-dd hh:mm:ss
##server.list server list, comma separated

```

```

##rf.enable.value Whether to enable rf synchronization {"false" disable "true" enable}
##rf.master.value Master {"false", "true"}
##rf.channel.value channel
-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="SetTimeInfo">
 <timezone value="##value"/>
 <summer enable="##enable"/>
 <sync value="##value"/>
 <time value="##value"/>
 <!--Added later, compatible with the old version-->
 <server list="##list"/>
 <rf>
 <enable value="##value"/>
 <master value="##value"/>
 <channel value="##value"/>
 </rf>
 </in>
</sdk>

<!--Feedback-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="7a8fe8bb88ccfab7dc57adc9c5f70e2e">
 <out result="kSuccess" method="SetTimeInfo"/>
</sdk>

```

# 18 Update function

## SDK firmware update

### ExcuteUpgradeShell firmware update

```

<!--
Firmware update request
##name is the firmware name, please make sure the firmware has been uploaded to the
device
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="##GUID">
 <in method="ExcuteUpgradeShell">
 <file name="##name"/>

```

```

 </in>
 </sdk>

 <!--
 Firmware update feedback-->
 <?xml version="1.0" encoding="utf-8"?>
 <sdk guid="##GUID">
 <out method="ExcuteUpgradeShell" result="kSuccess"/>
 </sdk>

```

## ExcuteUpgradeShellHttp Firmware update

ExcuteUpgradeShellHttp Support http to download firmware and then upgrade

```

 <!--
 Firmware Upgrade Request
 ##name is the firmware name or the path of http
 ##md5 The md5 value of the firmware
 ##size The size of the firmware
 -->
 <?xml version="1.0" encoding="utf-8"?>
 <sdk guid="##GUID">
 <in method="ExcuteUpgradeShellHttp">
 <file name="##name" md5="##md5" size="##size"/>
 </in>
 </sdk>

 <!--
 Firmware update feedback
 -->
 <?xml version="1.0" encoding="utf-8"?>
 <sdk guid="##GUID">
 <out method="ExcuteUpgradeShellHttp" result="kSuccess"/>
 </sdk>

```

## FirmwareUpgrade Firmware update

It is the same as ExcuteUpgradeShellHttp, but the status value returned in the download is different.

```

 <!--

```

Firmware Upgrade Request

##name is the firmware name or the path of http

##md5 The md5 value of the firmware

##size The size of the firmware

-->

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<sdk guid="##GUID">
```

```
 <in method="FirmwareUpgrade">
```

```
 <file name="##name" md5="##md5" size="##size"/>
```

```
 </in>
```

```
</sdk>
```

<!--

Firmware update feedback

-->

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<sdk guid="##GUID">
```

```
 <out method="FirmwareUpgrade" result="kSuccess"/>
```

```
</sdk>
```

## GetUpgradeResult Get firmware update status

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<sdk guid="##GUID">
```

```
 <in method="GetUpgradeResult"/>
```

```
</sdk>
```

<!--

Firmware upgrade feedback

##value is "success" (upgrade successful), "failed" (upgrade failed), "upgrading" (upgrading), "none" (no upgrade)

-->

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<sdk guid="##GUID">
```

```
 <out method="GetUpgradeResult" result="kSuccess">
```

```
 <upgrade result="##value"/>
```

```
 </out>
```

```
</sdk>
```



## 19 U-Disk function

### 19.1 Check the status of the U disk insert

**CheckUDiskInsert** Check whether the U disk is inserted

```
<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="CheckUDiskInsert"/>
</sdk>

<!-- Feedback
```

```

##value {"true" is connected, "false" is not connected}
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="7a8fe8bb88ccfab7dc57adc9c5f70e2e">
 <out result="kSuccess" method="CheckUDiskInsert">
 <insert value="##value"/>
 </out>
</sdk>

```

## 19.2 U-disk function configuration

### GetEnableUDiskFunction      Get      U-disk function configuration

```

<!--Request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="GetEnableUDiskFunction"/>
</sdk>

<!--feedback
##effective.value is allowed to be modified when the U disk is in what state {"all" all states
"insert" only when the U disk is connected}
##play.video and image must be the same, allowing to play videos and pictures in the root
directory of the U disk {"true" "false"}
##play.image and video must be the same
##import.config allow import configuration parameters {"true" "false"}
##import.immeProgram allows direct playback of U disk programs {"true" "false"}
##import.updateProgram allows to update U disk program {"true" "false"}
##import.upgrade allows to upgrade U disk firmware {"true" "false"}
##storage.enable Allow U disk storage {"true" "false"}
-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="7a8fe8bb88ccfab7dc57adc9c5f70e2e">
 <out result="kSuccess" method="GetEnableUDiskFunction">
 <effective value="##value"/>
 <play video="##video" image="##image"/>
 <import config="##config" immeProgram="##immeProgram"
updateProgram="##updateProgram" upgrade="##upgrade"/>
 <storage enable="##enable"/>
 </out>

```

```
</sdk>
```

## SetEnableUDiskFunction Enable or set U disk function configuration

```
<!--Request
##effective.value is allowed to be modified when the U disk is in what state {"all" all states
"insert" only when the U disk is connected}
##play.video and image must be the same, allowing to play videos and pictures in the root
directory of the U disk {"true" "false"}
##play.image and video must be the same
##import.config allow import configuration parameters {"true" "false"}
##import.immeProgram allows direct playback of U disk programs {"true" "false"}
##import.updateProgram allows to update U disk program {"true" "false"}
##import.upgrade allows to upgrade U disk firmware {"true" "false"}
##storage.enable Allow U disk storage {"true" "false"}
-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="SetEnableUDiskFunction">
 <effective value="##value"/>
 <play image="##image" video="##video"/>
 <import upgrade="##upgrade" updateProgram="##updateProgram"
config="##config" immeProgram="##immeProgram"/>
 <storage enable="##enable"/>
 </in>
</sdk>

<!--Feedback-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="7a8fe8bb88ccfab7dc57adc9c5f70e2e">
 <out result="kSuccess" method="SetEnableUDiskFunction"/>
</sdk>
```

## DisableUDiskFunction Close the U disk function configuration

When using SetEnableUDiskFunction, the U disk function configuration will be automatically enabled

```
<!--Request-->
```

```

<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="DisableUDiskFunction"/>
</sdk>

<!--Feedback-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="7a8fe8bb88ccfab7dc57adc9c5f70e2e">
 <out result="kSuccess" method="DisableUDiskFunction"/>
</sdk>

```

## 20. Expand function

# Custom Draw Analog Clock Properties

Color calculation formula

The value is color. It needs to be calculated and converted to a decimal value by itself

R=color & 255; red

G = (color >> 8) & 255; green

B = (color >> 16) & 255; blue

Turn off AM/PM related

```

<!-- 1 is displayed, the rest are closed -->

```

```
<clock Analog_AMPM_d="0">
...
</clock>
```

MinuteScaleType The MinuteScaleType attribute is the minute scale type 0 ellipse 1 rectangle 2 lines

The MinuteScaleColor property is the color for drawing the minute scale

HourHandColor hour hand color

MinuteHandColor minute hand color

SecondHandColor second hand color

```
<clock MinuteScaleColor="16711680" ...>
...
</clock>
```

## 21. Solutions to some problems

### 21.1 The data obtained by GetProgram is often used for program data.

#### Common black screen problem: incorrect resources tag.

In the program data obtained by GetProgram, the resource label is resource, and the resource label of adding a program and updating a program is resources with s. If the resource label is wrong, then this program will not exist. There are only regional programs. At this time, only The screen is black.

# The data of GetProgram

```
<!--This is data of request-->
<?xml version='1.0' encoding='utf-8'?>
<sdk guid="##GUID">
 <in method="GetProgram"/>
</sdk>

<!--This the data of return-->
<?xml version="1.0" encoding="utf-8"?>
<sdk guid="15811595b4daf72fe80e7e078de6edd4">
 <out result="kSuccess" method="GetProgram">
 <screen ...>
 <program ...>
 <playControl count="1"/>
 <border .../>
 <area ...>
 <rectangle height="32" x="0" y="0" width="32"/>
 <border .../>
 <!--When these data are used to update the data of the program, it
is the resource label that needs attention. This is without s, and the correct resource label
for updating data has s .resources this kind-->
 <resource>
 <text ...>
 <effect .../>
 <style .../>
 <string>test</string>
 <font size="20" name="Arial" italic="false"
underline="false" color="#ffff00" bold="true"/>
 </text>
 </resource>
 </area>
 </program>
 </screen>
 </out>
</sdk>
```

## 21.2

# Differences between HDPlayer programs and SDK programs

1. The text program drawn by HDPlayer is a picture drawn by HDPlayer, and then the program sent is a picture, not a text.
2. The program sent by the SDK is drawn by the device.
3. Because the drawing devices are different, the SDK program will need all the data to draw the corresponding program, and HDPlayer has already drawn it. So the program data does not contain all the xml data of the SDK. It cannot be used for SDK to update the program.

Conclusion: HDPlayer programs cannot be used as SDK programs. Therefore, SDK update programs cannot be used to update HDPlayer programs.

## Differences between HDPlayer and SDK programs.

1. HDPlayer programs are all drawn by themselves first. (So no drawing data is sent to the device, only program data is sent)
2. The SDK program sends the required data to the device. The device performs drawing processing. (All drawing data needs to be carried to the program data)

## Why doesn't HDPlayer send data to the device for processing like SDK does?

1. Poor performance. It is obvious that the performance of the computer and the equipment are poor. Drawing with a computer can be used to draw high-quality and delicate data.
2. Reduce the burden. When a program has a lot of content, depending on the device, some devices cannot draw all the data of the SDK. HDPlayer will draw all the data and then build the program list. At the same time, it will be drawn when sending. The data is sent down, so that the device does not need to draw again. The device does not need to bear the burden of drawing these data.

